Xin Li

Ph.D. Student

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https://lixin.ai
GitHub ▪ Google Scholar

Research

I work on **LLM agents**, as one loop. **Evaluate**: what can models do, where do they fail, and how reliably can we tell? **Train**: how can models learn from verifiers, rewards, and other models? **Orchestrate**: how should models, tools, and verifiers work together to complete tasks reliably? Each new way of orchestrating models creates failure modes that need to be evaluated in turn. So far mostly where an answer can be checked (formal verification, mathematics, code); now, self-improvement beyond checkable answers.

Education

2025 – 2029 (expected) **Ph.D., Nanyang Technological University**, Singapore
Electrical and Electronic Engineering. Advisor: Prof. Chau Yuen (IEEE Fellow)

2018 – 2021 **M.E., Peking University**
Electronics and Communication Engineering. Advisor: Prof. Jinlong Lin. Thesis: Leveraging Geometric Regularities for Robust Visual-Inertial Odometry

2014 – 2018 **B.E., Northeastern University, China**
Measurement and Control Technology

Selected Publications

Papers where I am first, joint first (*), or second author, grouped by the research loop (evaluate, train, orchestrate), then earlier work; newest first within each group. Full list on Google Scholar.

Evaluate

NeurIPS 2026 **WirelessMathBench-XL: An Auditable Benchmark for Wireless Mathematical Reasoning**
Xin Li, Mengbing Liu, Yiyang Zhu, Wenhe Zhang, Li Wei, Jiancheng An, Chau Yuen
[arXiv] [Project]

ICLR 2026 **Local Success Does Not Compose: Benchmarking Large Language Models for Compositional Formal Verification**
Xu Xu*, **Xin Li***, Xingwei Qu, Jie Fu, Binhang Yuan
[arXiv] [OpenReview] [Project]

Findings of ACL 2025 **WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications**
Xin Li, Mengbing Liu, Li Wei, Jiancheng An, Mérouane Debbah, Chau Yuen
[arXiv] [Project] [Code]

TMLR 2026 **RobustMAD: Evaluating Real-World Robustness of Multimodal Small Language Models for Deployable Anomaly Detection Assistants**
Anushiya Arunan, **Xin Li**, Yan Qin, U-Xuan Tan, Nhu Khue Vuong, Xiaoli Li, Chau Yuen
[OpenReview] [Project] [Code]

Train

arXiv 2026 **Beyond Teacher Assignment: Domain-Normalized Multi-Teacher On-Policy Distillation**
Xin Li, Hao Jiang, Xin Gao, Annan Wang, Yuchen Xie, Jinghao Guo, Xingwei Qu, Yichi Zhang, Chau Yuen
[arXiv] [Project] [Code]

TMLR 2026 **Re:Form—Reducing Human Priors in Scalable Formal Software Verification with RL in LLMs: A Preliminary Study on Dafny**
Chuanhao Yan*, Fengdi Che*, Xuhan Huang*, Xu Xu*, **Xin Li***, Yizhi Li*, Xingwei Qu*, Jingzhe Shi, Zhuangzhuang He, Chenghua Lin, Yaodong Yang, Binhang Yuan, Hang Zhao, Yu Qiao, Bowen Zhou, Jie Fu
[arXiv] [OpenReview] [Code]

npj Wireless Technology 2026 **Reconstructing wireless signals for low altitude networks using small language models**
Xin Li, Ran Liu, Chau Yuen
[Paper]

EMNLP 2026 Industry **GraphReduce: Coverage-Preserving LLM Aggregation for E-commerce Review Insights**
Hao Jiang, **Xin Li**, Yichi Zhang, Weisi Lin
[Project]

Orchestrate

NeurIPS 2026 **Measuring Collapse and Correction in Homogeneous-Panel LLM Debate**
Xin Li*, Mengbing Liu*, Chau Yuen
[Project]

Findings of EMNLP 2026 **Target-Local Verifier Choice in Best-of-K Reasoning Selection**
Xin Li, Hao Jiang, Weisi Lin
[Project] [Code]

AI4NextG @ NeurIPS 2025 (workshop) **LACP: LLM Agent Communication Protocol Requires Urgent Standardization**
Xin Li, Mengbing Liu, Chau Yuen
[arXiv] [Project]

Wireless systems

IEEE TSP 2026 **Stacked Intelligent Metasurface-Diffractive Deep Neural Networks for Onboard Terrain Classification from SAR Level-0 Raw Data**
Mengbing Liu, **Xin Li**, Jiancheng An, Chau Yuen
[Project] *extends our ML4RS @ ICLR 2025 workshop paper*

ICASSP 2025 **TransPathNet: A Novel Two-Stage Framework for Indoor Radio Map Prediction**
Xin Li, Ran Liu, Saihua Xu, Sirajudeen Gulam Razul, Chau Yuen
[arXiv] [Project] [Code]

Robot perception

IEEE RA-L 2020 **Co-Planar Parametrization for Stereo-SLAM and Visual-Inertial Odometry**
Xin Li*, Yanyan Li*, Evin Pınar Örnek, Jinlong Lin, Federico Tombari
[arXiv] [Code]

IROS 2020 **Leveraging Planar Regularities for Point-Line Visual-Inertial Odometry**
Xin Li*, Yijia He*, Jinlong Lin, Xiao Liu
[arXiv] [Code]

Experience

2024 – 2025 **Research Assistant**, *Nanyang Technological University*, Singapore.
Advisor: Prof. Chau Yuen. Built WirelessMathBench, the first benchmark for LLM mathematical modeling in wireless communications, and TransPathNet.

2022 – 2024 **SLAM Algorithm Engineer (Project Lead)**, *Gausium Robotics*, Singapore.
Led five engineers on hierarchical multimodal localization (vision, LiDAR, Wi-Fi) for commercial cleaning robots: 95% localization accuracy across 100,000+ m^2, deployed on a fleet of 1,000+ robots.

2020 – 2021 **Research Intern**, *Microsoft Research Asia*, Beijing.
Mentors: Dr. Yang Liu, Dr. Yizhong Zhang. RGB-D + IMU fusion for large-scale indoor mapping; vectorized maps of 10,000+ m^2 at sub-meter accuracy.

2019 – 2020 **Research Intern**, *MEGVII Technology*, Beijing.
Mentor: Dr. Yijia He. Real-time monocular visual-inertial odometry; semi-dense 3D mesh reconstruction at 30+ FPS.

Grants and Awards

2026 NeurIPS 2026 Top Reviewer
2026 Google Gemini Academic Program Award (US$10,000)
2025 Modal Academics Compute Grant (US$2,000)
2025 Cohere Labs Catalyst Grant (US$1,500)
2025 OpenAI Researcher Access Program (US$1,000)
2025 Rohde & Schwarz Award, IEEE 6G Summit Singapore
2025 PREMIA Best Student Paper Award, finalist
2025 NTU Research Scholarship (full Ph.D. funding)

Mentoring

2026 **Chengqi Liang**, M.Sc. dissertation, NTU; went on to a Ph.D. at CUHK-Shenzhen

2026 **Yukun Jin**, B.Eng. final-year project, NTU (Wuhan University 3.5+0.5+1 programme); went on to an M.Sc. at NTU

2024 **Haoyu Xu**, M.Comp. dissertation, National University of Singapore; went on to a Ph.D. at Peking University

Invited Talks

Oct. 2025 **Teaching Large Language Models Mathematical Reasoning in Wireless Communications: From Benchmarking to Efficient Training**
ISEE Youth Scholar Forum, School of Information and Electronic Engineering, Zhejiang University

Jun. 2025 **WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications**
NICE Session 66: Evaluation Paper Sharing @ ICML & ACL 2025

Service

Organizer AIR4D @ IROS 2025, Workshop on Advancements for Intelligent Robotics in 4D Scenes

Reviewer NeurIPS (Top Reviewer, 2026), ICLR, ICML, AAAI, AISTATS, CVPR, ECCV, SIGGRAPH, ICRA, IROS; IEEE RA-L, ACM TOG, IEEE TNNLS